INITIAL DRAFT · 29 OCTOBER 2026 SGT

Different routines.
One common ground.

FlatSplit helps 2–5 flatmates compare Singapore towns without hiding one person behind the group average.

CS5224 project · All six authors review before submission · This is not a final deployed-build attestation.

THE PROBLEM

An affordable average can still mean an unfair journey.

Budget

Each flatmate has a personal rent limit.

Commute

Different destinations produce different journeys.

Visibility

Show individual outcomes before the group chooses.

Scope: town-level estimates, equal whole-flat rent split, public transport. No listings, availability guarantee, booking or payments. Source: user-journey.md and API contract.

WORKING JOURNEY

Share. Save. Search. Compare.

1 / Shared inputs

Create a group, privately share an invite, then each member saves their own destination and limits.

2 / Async search

Read actual progress. Preserve inputs during errors. Hide results after preferences change.

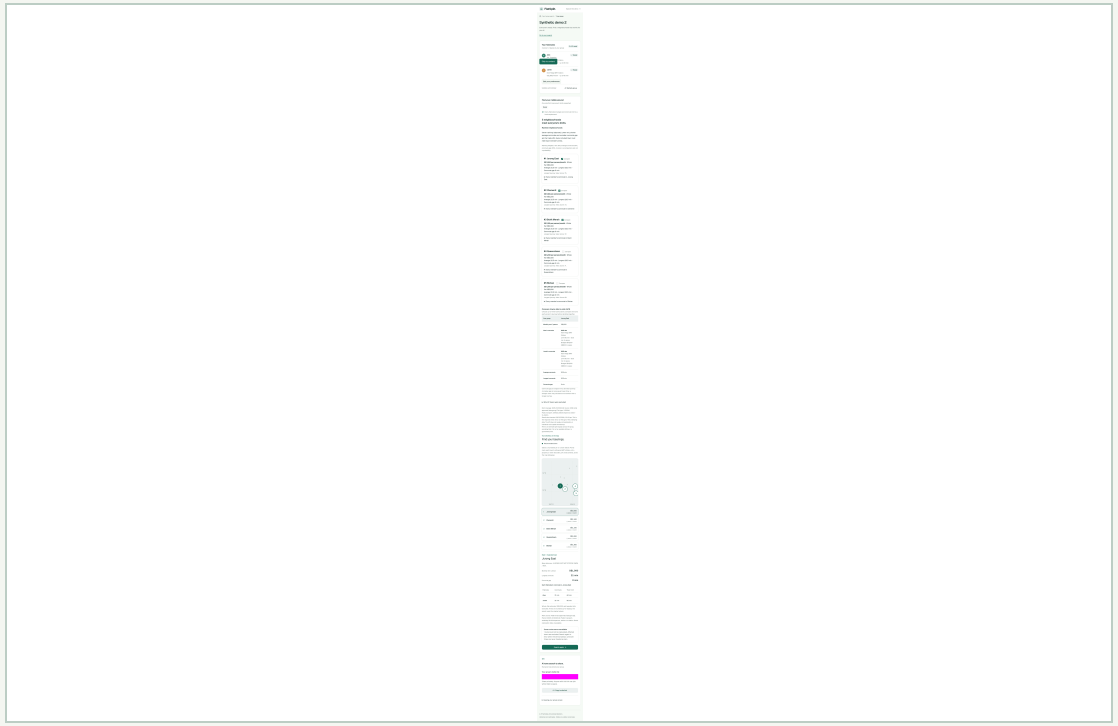
3 / Explain trade-offs

Compare three towns, every member’s commute, rent/person and the longest-to-shortest gap.

Implementation artifacts: #84 / #85 and the local integration preparation of #76 / #77. Code and fixture QA are separate from deployed acceptance.

LOCAL UI EVIDENCE — SYNTHETIC DATA

The comparison stays usable on a phone.

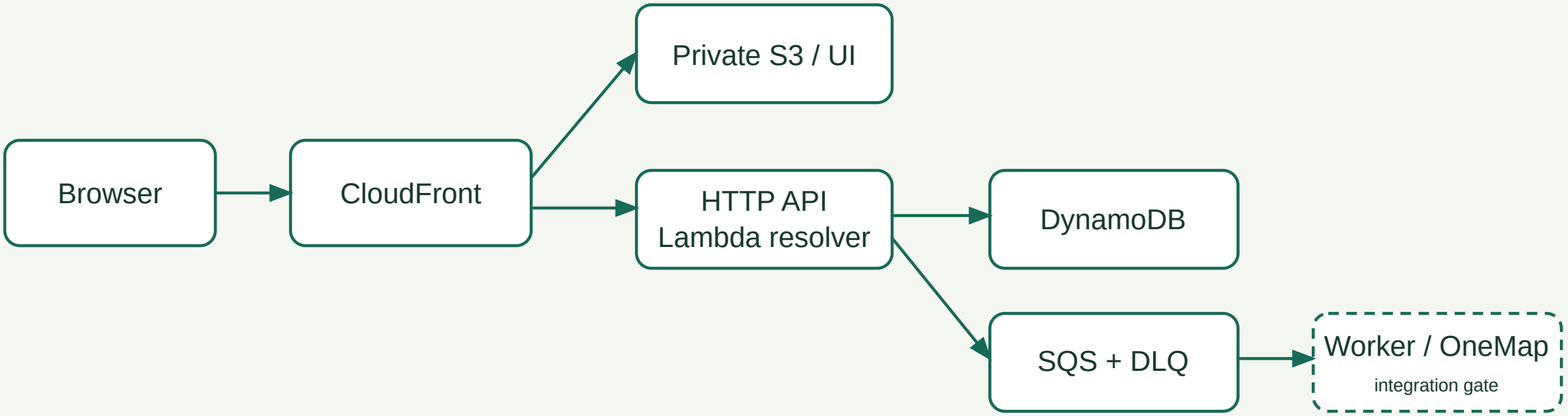


- 390px group input, results and map.
- Keyboard selection and a scrollable three-town table.
- Per-member editing survives independent browser reloads.

Moto + real local HTTP resolver. Synthetic places/rent and injected conditional route completions. Replace this image with evidence from the final deployed SHA before submission.

SOURCE-DEFINED AWS ARCHITECTURE

One request path; routing is asynchronous.



Solid boxes: code-defined topology at main b4a4434; not a current runtime inventory. #74 worker and #79 alarms are reviewed PR work, not silently included in that main. Secrets remain server-side. Source: infra/aws/src/{web,api,data}.ts and #39 backend packet.

CORRECTNESS BEFORE RECOMMENDATIONS

Retries and edits must not make old data look current.

- Invite capability joins; member capability reads and edits its own preferences.
- Searches pin the group revision and data inputs; changed preferences withhold old results.
- Unknown route times are unavailable, never zero. A timeout cannot prove a preference write did not commit.

#67 privacy corrections independently verified at 92a480c: 32 focused tests + 11 local scenarios / 175 calls, as reported in the dated review. This supersedes the earlier “privacy finding open” packet; it is controlled local proof, not deployed/browser acceptance.

Source: API contract, #67 review handback, #39 correction ledger

DATA AND FAIRNESS

Be explicit about the estimate.

Rent

Historical whole-flat summaries, equal split, uncertainty and coverage shown. Not an available listing.

Commute

Station-to-station estimate for a specified travel basis. Snapping and provider availability limit precision.

Ranking

Filter every member’s hard limits, then show rent, average commute and commute-gap trade-offs.

Current SearchView lacks rent-publication and per-route update timestamps. “Last response checked” is not data freshness. #75’s latest timestamp correction still needs formal acceptance; no accepted real publication is established by this deck.

EVALUATION: THREE DIFFERENT KINDS OF STATEMENT

Targets are not measurements.

Statement	What we can say now	What it does not prove
Local correctness	Versioned tests and synthetic HTTP/browser QA; see source ledger.	Real provider, cloud release or participant acceptance.
Warm result latency	p95 ≤ 500 ms is the agreed target.	No accepted measured p95 or maximum throughput is claimed.
AWS cost	Scenario models and operation-count assumptions exist.	A model is not an AWS bill or measured consumed capacity.
User study	#78 protocol/cards prepared; minimum three eligible participants, target 8–12.	No participant session or usability gain is claimed.

#80 interpreter reproducibility is a numerical correction, not a new cost measurement. Drain arithmetic such as 78 pairs/~28s is extrapolation, never measured throughput.

DEMO AND RECOVERY

Show what happens when plans change.

1. Create/share with two independent member contexts.
2. Save explicit destinations; show pending and compare three actual returned towns.
3. Edit a budget, reload the other member, then show stale-result withholding.
4. Explain missing rent/routes and safe retry without inventing a completed recommendation.

Use docs/demo/README.md. Local synthetic mode is reproducible now; final demonstration needs the reviewed deployed SHA, known sample inputs, real data/provenance and redacted screenshots from that same build. Never disrupt the shared deployment for a failure demo.

TEAM ROLES — ESTABLISHED TASK OWNERSHIP

Six contributions; one reviewed story.

Member	Implementation/evidence contribution	Proposed presentation handoff
Raccoon33 / BBQQ	Forms, comparisons, mobile pass, demo, slide coordination	Problem → live journey → final limitations
Hikaru	Journey, sharing, status, map, recovery, study	UX choices and user-study limits
Mani	Rental pipeline, quality and report coordination	Data lineage and uncertainty
Ming	Lookup/worker, operational recovery and cost model	Async routing and modeled costs
WK	Persistence/search, correctness, performance/security evidence	Backend invariants and measured-evidence gaps
Shyam	Infrastructure, API contracts, filtering/ranking, evaluation synthesis	AWS design and evaluation interpretation

Speaking order/durations are proposals awaiting all six members' agreement, not assigned or completed rehearsal.

Source: ownership tables in #41 / #44

CONCLUSION AND REMAINING GATES

A shortlist is useful only when every flatmate’s outcome is visible.

- Local preparation now connects member edits, comparisons, map and recovery.
- Final claims need reviewed deployment, real rent/route evidence and participant observations.
- All-six-member review and a timed rehearsal precede submission.

Final report and slides:

13 November 2026, 23:59 Singapore time.

CITATION LEDGER AND FINAL CHECKS

Carry the limits with each claim.

- API contract / backend design / evaluation plan.
- #39 source packet and later corrections; #67 92a480c supersedes earlier privacy warnings; #74/#79 clean heads are not obsolete conflicts.
- Demo procedure, mobile/accessibility pass and remaining-work matrix.
- Before finalisation: replace synthetic screenshots, reconcile every slide with the report/deployed SHA, add actual measurements, record all six approvals, timed rehearsal and submission receipt.

Use the detailed speaker notes and citation ledger in docs/presentation/README.md. This deck does not imply all-six review, a live study, successful deployment or completed submission.